

ASKEY-DECORRELATED AND PRECONDITIONED LEAST SQUARES VARIATIONAL INFERENCE

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ABSTRACT

Least Squares Variational Inference (LSVI) converts unnormalised Kullback-Leibler (uKL) divergence minimisation into a sequence of ordinary least squares (OLS) regressions, enabling gradient-free natural gradient steps. However, standard LSVI suffers from an $\mathcal{O}(m^3 + m^2N)$ computational complexity per iteration due to empirical Fisher Information Matrix (FIM) inversion, as well as catastrophic numerical instability and parameter divergence when the Monte Carlo sample size N is small relative to the statistic dimension m ($N \ll m$). In this work, we present **Askey-Decorrelated and Preconditioned LSVI (AD-LSVI)**, a scalable framework that resolves these computational and statistical bottlenecks across arbitrary exponential families through four integrated components: (1) an Askey orthonormal polynomial basis $t_\eta(x)$ that decouples empirical FIMs without matrix inversions; (2) a dual-space parameter mapping matrix M_η enabling inversion-free $\mathcal{O}(mN)$ natural parameter updates; (3) a Stein Block-Jacobi Preconditioned Conjugate Gradient (PCG) solver with residual-tolerance stopping for custom non-Askey families; and (4) variance-reduced score-function control variates with adaptive temporal smoothing and scale-free residual caps. Across four benchmark logistic regression targets (Pima, Sonar, MNIST, Census Income) and non-Gaussian classical families (Gamma, Beta, Poisson), AD-LSVI eliminates small- N divergences (0/50 vs 10/50 on Pima at $N = 200$), cuts target update variance by up to $6\times$, reduces per-step runtime by $\sim 3\times$, and maintains robust convergence in extreme sample-limited regimes ($N \ll m$) where standard LSVI and Natural Gradient Descent collapse.

1 INTRODUCTION

Variational inference (VI) Blei et al. (2016); Hoffman et al. (2012) has emerged as a cornerstone of modern Bayesian computation, recasting intractable posterior integration into a scalable optimization problem. Standard approaches—such as Automatic Differentiation Variational Inference (ADVI) Kucukelbir et al. (2016), Black-Box Variational Inference (BBVI) Ranganath et al. (2013), and Auto-Encoding Variational Bayes Kingma & Welling (2013)—typically optimize the Evidence Lower Bound (ELBO) via first-order stochastic gradient methods like Adam Kingma & Ba (2014). However, standard gradient descent in Euclidean space often exhibits slow convergence on ill-conditioned objectives because it ignores the intrinsic geometric structure of the variational family Raskutti & Mukherjee (2013). To address this limitation, Natural Gradient Descent (NGD) Hoffman et al. (2012); Martens (2014); Zhang et al. (2017) and Mirror Descent (MD) Khan & Lin (2017); Hanzely & Richtárik (2018); Tomar et al. (2020) incorporate Riemannian curvature via the Fisher Information Matrix (FIM), achieving faster convergence by taking steepest descent steps along the information manifold.

Recently, Least Squares Variational Inference (LSVI) Fay et al. (2025); Salimans & Knowles (2012; 2014) has re-framed variational updates for unnormalised Kullback-Leibler (uKL) minimization Matthews et al. (2015); Li et al. (2015) as a sequence of ordinary least squares (OLS) regressions. In exponential families, the first-order optimality condition of uKL corresponds to a fixed-point equation whose iteration performs an exact natural gradient step in the space of natural parameters and a mirror descent step in the dual space of moment parameters Fay et al. (2025); Lin et al. (2019). Crucially, LSVI computes updates directly from Monte Carlo evaluations of the target log-density

without requiring explicit gradients of the target, making it well-suited for non-differentiable or likelihood-free objectives Fay et al. (2025).

Despite these theoretical advantages, existing LSVI formulations suffer from severe computational and statistical bottlenecks in practical, high-dimensional applications:

1. **High Computational Complexity:** Generic LSVI inverts the empirical Fisher matrix $\hat{F}_\eta = \frac{1}{N} \sum_{i=1}^N s(X_i)s(X_i)^\top$ ($s(x) \in \mathbb{R}^m$) each step, an $\mathcal{O}(m^3 + m^2N)$ cost that is prohibitive for moderately large m Mishkin et al. (2018).
2. **Empirical FIM Singularity and Variance Explosion:** When $N \ll m$, $\hat{F}_\eta$ becomes rank-deficient or singular Kunstner et al. (2019); Pasand & Dohmatob (2025); Wu et al. (2024), and inverting it injects extreme variance that destabilizes or diverges the optimization.
3. **Restriction to Gaussian Families:** Prior reparameterization transforms ($t(Z)$) bypassing inversion Fay et al. (2025) were tailored to Gaussian families; general non-Gaussian families (Gamma, Beta, Poisson, Dirichlet, von Mises-Fisher) remained an open challenge.
4. **Target Evaluation Variance:** Monte Carlo feature-target projections incur high variance for noisy or heavy-tailed $f(x)$, degrading convergence Geffner & Domke (2020); Luu et al. (2024); Roeder et al. (2017).

To overcome these structural limitations, we introduce **Askey-Decorrelated and Preconditioned LSVI (AD-LSVI)**, a unified, sample-efficient, and computationally scalable framework for variational inference across arbitrary classical and custom exponential families. AD-LSVI addresses the core bottlenecks of LSVI through four tightly integrated mathematical contributions:

- **Askey Orthonormal Decorrelation Basis $t_\eta(x)$ (Component 1):** For classical families we build orthonormal bases (Laguerre/Gamma, Jacobi/Beta, Charlier/Poisson) via the Wiener-Askey scheme, satisfying $\mathbb{E}_\eta[t_\eta t_\eta^\top] = I_m$ and evaluated in $\mathcal{O}(mN)$ by 3-term recurrences without forming or inverting covariance matrices.
- **Dual-Space Parameter Mapping M_η (Component 2):** A lower-triangular change-of-basis $t_\eta(x) = M_\eta s(x)$ yields the exact identity $F_\eta^{-1} = M_\eta^\top M_\eta$, enabling inversion-free updates $\hat{\eta}'_{t+1} = M_\eta^\top \hat{\gamma}_t$ in $\mathcal{O}(mN)$ time with a guaranteed identity effective FIM for *any* budget $N \geq 1$, resolving small- N singularity.
- **Correlation-Aware Block-Jacobi PCG Solver (Component 3):** For custom families lacking analytical bases, we solve $F_{\eta_t} \hat{\eta}'_{t+1} = \hat{z}_{\eta_t}$ via matrix-free Preconditioned CG with a block-Jacobi preconditioner and residual-tolerance stopping (10^{-8}), avoiding dense $\mathcal{O}(m^3)$ inversion for large m and the under-solving artifacts of diagonal Jacobi or fixed K -truncation.
- **Score-Function Control Variates with Smoothing (Component 4):** Zero-mean score control variates $g_\eta(x) = s(x) - \nabla A(\eta)$ in decorrelated coordinates, with temporal smoothing (EMA/burn-in) and scale-free residual caps, halve update variance under $m > N$.

Across logistic regression posteriors (Pima, Sonar, MNIST, Census Income) and non-Gaussian classical families (Gamma, Beta, Poisson), AD-LSVI eliminates small- N divergence, improves sample efficiency (optimal ELBOs with up to $6\times$ fewer samples than standard LSVI), and reduces parameter estimator variance by orders of magnitude, all while preserving the derivative-free nature of least-squares updates.

2 RELATED WORK

Our proposed AD-LSVI framework connects three distinct research paradigms in approximate inference and optimization: natural gradient variational inference, stochastic variance reduction with control variates, and matrix-free curvature solvers.

2.1 NATURAL GRADIENT DESCENT & LEAST SQUARES VARIATIONAL INFERENCE

Natural Gradient Descent (NGD) Martens (2014); Ollivier (2013); Raskutti & Mukherjee (2013) pre-conditions updates by the Fisher Information Matrix (FIM); in exponential-family VI it is formally equivalent to Mirror Descent in the dual moment space Khan & Lin (2017); Hanzely & Richtárik (2018); Tomar et al. (2020); Lin et al. (2019). Despite appealing convergence guaran-

tees Ding et al. (2022), NGD requires inverting the FIM each iteration, an $\mathcal{O}(m^3)$ bottleneck in the statistic dimension m Zhang et al. (2017); Khan et al. (2018). Structured approximations mitigate this cost via Kronecker factorization (KFAC) Martens & Grosse (2015); Grosse & Martens (2016); Dangel et al. (2024), low-rank-plus-diagonal Mishkin et al. (2018), Kronecker eigenbases George et al. (2018), or block-diagonal Gauss-Newton structure Botev et al. (2017). When the FIM is estimated from few samples ($N \ll m$), however, the empirical Fisher matrix becomes ill-conditioned or singular Kunstner et al. (2019); Pasand & Dohmatob (2025), injecting destructive noise into updates Wu et al. (2024) unless explicit ridge damping or shrinkage is applied Jin et al. (2019; 2018).

Least Squares Variational Inference (LSVI) Fay et al. (2025), building on stochastic linear-regression formulations Salimans & Knowles (2012; 2014), recasts unnormalised KL minimization Matthews et al. (2015); Li et al. (2015) as ordinary least squares. It bypasses target gradients but still inverts the empirical Gram matrix $\hat{F}_\eta$, inheriting $\mathcal{O}(m^3)$ cost and rank deficiency when $N < m$. Le Fay et al. (2025) give tailored $\mathcal{O}(d^3)/\mathcal{O}(d)$ updates for full-covariance and mean-field Gaussians via $t(Z)$ reparameterization, but these do not extend to non-Gaussian families. AD-LSVI instead uses Askey orthonormal bases and dual-space mapping matrices M_η for inversion-free $\mathcal{O}(mN)$ updates with a guaranteed identity effective FIM across arbitrary classical families (Gamma, Beta, Poisson).

2.2 STOCHASTIC VARIATIONAL INFERENCE & CONTROL VARIATES

Stochastic Variational Inference (SVI) Hoffman et al. (2012); Blei et al. (2016) scales approximate Bayesian inference to large datasets using stochastic gradient optimization. Black-Box Variational Inference (BBVI) Ranganath et al. (2013) computes score-function (REINFORCE) gradients, which apply to general non-differentiable target densities but suffer from high Monte Carlo variance. Automatic Differentiation Variational Inference (ADVI) Kucukelbir et al. (2016) and Variational Autoencoders Kingma & Welling (2013) leverage pathwise reparameterization gradients to achieve lower variance for continuous, differentiable targets Mohamed et al. (2019).

To tame estimator variance, control variates and variance-reduction techniques have been widely studied Paisley et al. (2012); Mnih & Rezende (2016); Roeder et al. (2017), including continuous relaxations (REBAR Tucker et al. (2017), RELAX Grathwohl et al. (2017)), log-variance losses (VarGrad Richter et al. (2020)), Quasi-Monte Carlo Buchholz et al. (2018), and optimal-transport flows Luu et al. (2024). Estimating high-dimensional control-variate weights Θ_t afresh from small mini-batches ($m > N$), however, introduces severe second-order sampling noise Miller et al. (2017); Geffner & Domke (2018); Domke (2019) that can destabilize trajectories or distort covariance estimation Geffner & Domke (2020); Fan et al. (2015). AD-LSVI constructs zero-mean score control variates $g_\eta(x) = \nabla_\eta \log q_\eta(x)$ in decorrelated coordinates, combined with temporal smoothing (EMA or burn-in freeze) and residual caps to stabilize updates under tight budgets.

2.3 CURVATURE MATRIX INVERSION & PRECONDITIONED LINEAR SOLVERS

Solving the curvature system $F_\eta \Delta\eta = z_\eta$ without forming dense $m \times m$ matrices is a central challenge in second-order optimization Dauphin et al. (2015); Pascanu & Bengio (2013). Methods such as TRPO Schulman et al. (2015) and ACKTR Wu et al. (2017) use Truncated Conjugate Gradient (tCG) with matrix-free Fisher-vector products. Standard diagonal (Jacobi) preconditioning fails when ill-conditioning stems from off-diagonal correlations rather than diagonal scaling Dauphin et al. (2015); Li (2022), and truncating CG at a fixed count (e.g., $K = 20$) under-solves ill-conditioned systems, causing fidelity errors and artificial entropy over-dispersion Müller & Zeinhofer (2023). For custom families lacking analytical Askey bases, AD-LSVI employs a correlation-aware block-Jacobi preconditioner with residual-tolerance stopping ($\text{tol} = 10^{-8}$), preserving trajectory fidelity while reserving matrix-free solves for high-dimensional regimes ($m \geq 4096$) where direct Cholesky factorization is bottlenecked. First-order optimizers such as Adam Kingma & Ba (2014) serve as standard black-box baselines.

3 METHODOLOGY

We present the mathematical framework of **Askey-Decorrelated and Preconditioned LSVI (AD-LSVI)**. An architectural schematic illustrating its execution tracks is provided in Figure 1.

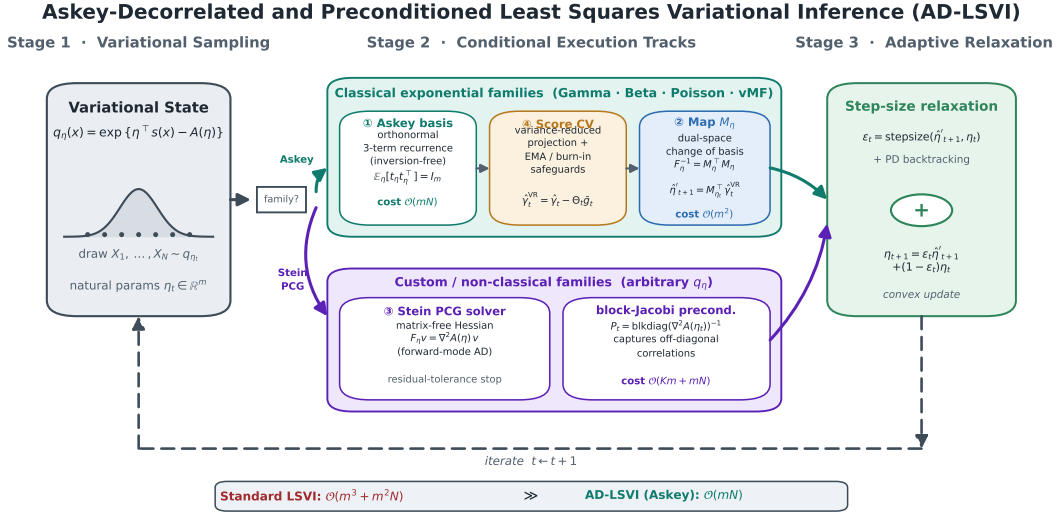


Figure 1: Architectural workflow of Askey-Decorrelated and Preconditioned Least Squares Variational Inference (AD-LSVI). The pipeline begins with variational sampling and branches into two conditional execution paths. Classical exponential families are processed via Askey orthonormal bases evaluated using three-term recurrence relations, variance-reduced score control variates with temporal exponential moving average safeguards, and dual-space matrix transformations for inversion-free natural parameter updates. Custom non-classical families are updated using a matrix-free Stein preconditioned conjugate gradient solver with correlation-aware preconditioning and dynamic residual safeguards. Candidate parameter updates from either branch are integrated using adaptive step-size relaxation before feeding back into the iterative sampling loop.

3.1 THEORETICAL FOUNDATIONS OF LSVI & UNNORMALISED KL DIVERGENCE

Let $\pi(x) = \exp f(x)$ be an unnormalised target density on $\mathcal{X} \subseteq \mathbb{R}^d$, where $f(x) = \log \pi(x)$ is the target log-density. We approximate $\pi(x)$ using a regular, minimal exponential family $Q = \{q_\eta : \eta \in \mathcal{V}\}$, parameterized by natural parameters $\eta \in \mathcal{V} \subseteq \mathbb{R}^m$:

$$q_\eta(x) = \exp\{\eta^\top s(x) - A(\eta)\}, \quad (1)$$

where $s(x) = (1, \bar{s}(x)^\top)^\top \in \mathbb{R}^m$ is the extended vector of linearly independent sufficient statistics with an explicit intercept term, $A(\eta) = \log \int_{\mathcal{X}} \exp\{\eta^\top s(x)\} dx$ is the log-partition function, and $Z_\eta = \exp A(\eta)$ is the partition function.

Variational inference minimizes the unnormalised Kullback-Leibler (uKL) divergence Matthews et al. (2015); Li et al. (2015); Fay et al. (2025):

$$\text{uKL}(q_\eta | \pi) := \int_{\mathcal{X}} q_\eta(x) \log \left(\frac{q_\eta(x)}{\pi(x)} \right) dx + Z(\pi) - Z_\eta. \quad (2)$$

Minimizing $\text{uKL}(q_\eta | \pi)$ with respect to η yields the first-order stationarity condition:

$$\nabla_\eta \text{uKL}(q_\eta | \pi) = 0 \iff F_\eta \eta = z_\eta, \quad (3)$$

where $F_\eta := \mathbb{E}_\eta[s(X)s(X)^\top] = \nabla^2 A(\eta) + \nabla A(\eta) \nabla A(\eta)^\top$ is the Fisher Information Matrix (FIM) appended with raw second moments, and $z_\eta := \mathbb{E}_\eta[f(X)s(X)]$ is the target-feature cross-moment vector under $X \sim q_\eta$.

The exact LSVI mapping $\phi(\eta) := F_\eta^{-1} z_\eta$ corresponds to the Ordinary Least Squares (OLS) regressor of $f(X)$ onto $s(X)$ under $X \sim q_\eta$:

$$\phi(\eta) = \arg \min_{\beta \in \mathbb{R}^m} \mathbb{E}_\eta \left[(\beta^\top s(X) - f(X))^2 \right]. \quad (4)$$

To ensure stability, exact LSVI employs momentum fixed-point updates with step sizes $\varepsilon_t \in (0, 1]$:

$$\eta_{t+1} = \varepsilon_t \phi(\eta_t) + (1 - \varepsilon_t) \eta_t = \eta_t - \varepsilon_t F_{\eta_t}^{-1} \frac{\nabla_{\eta} \text{uKL}(q_{\eta_t} | \pi)}{Z_{\eta_t}}, \quad (5)$$

which is formally equivalent to a Natural Gradient Descent step on $\text{uKL}(q_{\eta} | \pi)$ in natural parameter space, and Mirror Descent in moment space $\omega(\eta) = \nabla A(\eta)$ Raskutti & Mukherjee (2013); Fay et al. (2025).

Step-Size Schedule & Trust-Region Safeguards: In standard LSVI, the step size follows a decreasing harmonic schedule $\varepsilon_t = 1/(t+1)$, ensuring asymptotic convergence under Robbins-Monro stochastic approximation conditions Fay et al. (2025). In practice, fixed step sizes $\varepsilon_t = \varepsilon_0$ or adaptive line-search bounds can be integrated seamlessly. To prevent step overshooting and maintain positive-definiteness of the variational family, AD-LSVI incorporates a natural-gradient step-backtracking safeguard: if a candidate update η_{t+1} violates parameter domain constraints $\mathcal{V}$ or increases target KL divergence catastrophically, the step size is iteratively halved ($\varepsilon_t \leftarrow \varepsilon_t/2$).

3.2 COMPONENTS 1 & 2: ASKEY DECORRELATION BASIS & PARAMETER MAPPING M_{η}

In generic LSVI, evaluating $\phi(\eta)$ via empirical sample averages $\hat{F}_{\eta} = \frac{1}{N} \sum_{i=1}^N s(X_i) s(X_i)^{\top}$ requires inverting an empirical $m \times m$ Gram matrix. This incurs an $\mathcal{O}(m^3 + m^2 N)$ computational bottleneck per iteration and produces singular matrices when $N < m$ Kunstner et al. (2019).

To eliminate matrix inversion entirely, we exploit the Wiener-Askey scheme of orthogonal polynomials. For any regular classical exponential family $q_{\eta}(dx)$, there exists an orthonormal polynomial basis vector $t_{\eta}(x) = (t_0(x), t_1(x), \dots, t_{m-1}(x))^{\top} \in \mathbb{R}^m$ associated with the probability measure $q_{\eta}(dx)$ satisfying exact orthogonality:

$$\mathbb{E}_{\eta} [t_{\eta}(X) t_{\eta}(X)^{\top}] = I_m. \quad (6)$$

Classical Askey Families:

- **Gamma Family $\text{Gamma}(\alpha, \beta)$:** Standardized Generalized Laguerre polynomials $t_k(x) = \tilde{L}_k^{(\alpha-1)}(\beta x)$.
- **Beta Family $\text{Beta}(\alpha, \beta)$:** Standardized Jacobi polynomials $t_k(x) = \tilde{P}_k^{(\alpha-1, \beta-1)}(2x-1)$.
- **Poisson Family $\text{Poisson}(\lambda)$:** Standardized Charlier polynomials $t_k(x) = \tilde{C}_k(x; \lambda)$.
- **Gaussian / von Mises-Fisher Family:** Standardized Hermite or Gegenbauer polynomials.

Evaluating orthonormal features $t_k(X_i)$ for N samples $X_1, \dots, X_N \sim q_{\eta_t}$ is executed in $\mathcal{O}(mN)$ time via 3-term recurrence relations:

$$t_{k+1}(x) = (A_k x + B_k) t_k(x) - C_k t_{k-1}(x), \quad k = 1, \dots, m-2, \quad (7)$$

where A_k, B_k, C_k are analytical scalar functions of the parameter η_t .

Dual-Space Mapping Matrix M_{η} : Because $s(x)$ and $t_{\eta}(x)$ span the same degree- d polynomial subspace, there exists a unique lower-triangular matrix $M_{\eta} \in \mathbb{R}^{m \times m}$ such that:

$$t_{\eta}(x) = M_{\eta} s(x). \quad (8)$$

By orthonormality, $\mathbb{E}_{\eta} [t_{\eta}(X) t_{\eta}(X)^{\top}] = M_{\eta} \mathbb{E}_{\eta} [s(X) s(X)^{\top}] M_{\eta}^{\top} = M_{\eta} F_{\eta} M_{\eta}^{\top} = I_m$. Thus, the exact inverse FIM decomposes analytically without matrix inversion:

$$F_{\eta}^{-1} = M_{\eta}^{\top} M_{\eta}. \quad (9)$$

Let $\hat{\gamma}_t := \frac{1}{N} \sum_{i=1}^N t_{\eta_t}(X_i) f(X_i)$ be the Monte Carlo projection of target values onto decorrelated features. The candidate target natural parameter update $\hat{\eta}'_{t+1}$ is computed directly in $\mathcal{O}(m^2)$ time (or $\mathcal{O}(mN)$ for banded M_{η}):

$$\hat{\eta}'_{t+1} = M_{\eta_t}^{\top} \hat{\gamma}_t. \quad (10)$$

Because $\mathbb{E}_{\eta} [t_{\eta} t_{\eta}^{\top}] = I_m$ holds analytically, the effective Fisher conditioning is exactly 1.0 for *any* sample size $N \geq 1$, completely resolving small- N singularity.

Extension to Coupled Multivariate & Non-Factorized Families: For factorized products $q_{\eta}(x) = \prod_{j=1}^d q_{\eta_j}(x_j)$, the basis $t_{\eta}(x)$ is a tensor product of univariate Askey polynomials over the

marginals. For coupled families with non-diagonal covariance (where analytical multivariate bases are intractable), AD-LSVI falls back to Component 3, which updates matrix-free via the log-partition Hessian $F_\eta = \nabla^2 A(\eta)$ regardless of coupling.

3.3 COMPONENT 3: STEIN-PRECONDITIONED BLOCK-JACOBI SOLVER FOR CUSTOM FAMILIES

For custom exponential families where explicit Askey orthonormal bases are unavailable, the exact FIM is available in closed form via the log-partition Hessian: $F_\eta = \nabla^2 A(\eta) = \text{Cov}_\eta(s(X))$. To avoid forming or inverting dense $m \times m$ empirical matrices at $\mathcal{O}(m^3)$ cost, we compute the update $\hat{\eta}'_{t+1}$ by solving $F_{\eta_t} \hat{\eta}'_{t+1} = \hat{z}_{\eta_t}$ using Preconditioned Conjugate Gradient (PCG).

Standard Jacobi (diagonal) preconditioning $P_t = \text{diag}(\nabla^2 A(\eta_t))^{-1}$ fails when target ill-conditioning is driven by off-diagonal quadratic correlations Dauphin et al. (2015). To resolve this, we construct a correlation-aware **Block-Jacobi preconditioner**:

$$P_t = \text{block-diag}(\nabla^2 A(\eta_t))^{-1}, \quad (11)$$

where blocks correspond to logical parameter groupings (e.g., mean vector vs. covariance blocks). Matrix-vector products $F_{\eta_t} v = \nabla^2 A(\eta_t) v$ are evaluated matrix-free in $\mathcal{O}(m)$ time via forward-mode automatic differentiation.

Block Partitioning Sensitivity & Fallback Rules: Absent an explicit mean/covariance separation, blocks default to fixed $b \times b$ structural partitioning ($b \in \{16, 32\}$) or coordinate clustering from the non-zero pattern of $\nabla^2 A(\eta)$. Even under arbitrary block choices, block-Jacobi robustly captures local off-diagonal correlations, giving order-of-magnitude acceleration over standard Jacobi.

To eliminate under-solving artifacts and covariance entropy distortion caused by fixed truncation ($K = 20$), PCG iterations terminate strictly based on a relative residual tolerance:

$$\frac{\|F_{\eta_t} p_k - \hat{z}_{\eta_t}\|_2}{\|\hat{z}_{\eta_t}\|_2} < 10^{-8}. \quad (12)$$

Computational Scaling & Crossover Regime: As established empirically in Section 4.4, direct dense Cholesky (BLAS-3) is faster for $m \leq 2048$, whereas matrix-free Block-PCG dominates for $m \geq 4096$, reducing per-step solver memory by up to $101\times$ and runtime by $> 14\times$.

3.4 COMPONENT 4: VARIANCE-REDUCED SCORE CONTROL VARIATES & SAFEGUARDS

Monte Carlo projections $\hat{\gamma}_t$ exhibit sample noise that can destabilize natural gradient updates. We construct a zero-mean score control variate vector using the log-derivative of $q_\eta(x)$:

$$g_\eta(x) = \nabla_\eta \log q_\eta(x) = s(x) - \nabla A(\eta), \quad \text{with} \quad \mathbb{E}_\eta[g_\eta(X)] = 0. \quad (13)$$

In decorrelated coordinates, the variance-reduced projection $\hat{\gamma}_t^{\text{VR}}$ is computed as:

$$\hat{\gamma}_t^{\text{VR}} = \frac{1}{N} \sum_{i=1}^N (t_{\eta_t}(X_i) f(X_i) - \Theta_t g_{\eta_t}(X_i)), \quad (14)$$

where $\Theta_t = \text{diag}(\theta_{t,1}, \dots, \theta_{t,m})$ is a diagonal weight matrix updated in closed form:

$$\theta_{t,j} = \frac{\widehat{\text{Cov}}(t_{\eta_t,j}(X) f(X), g_{\eta_t,j}(X))}{\widehat{\text{Var}}(g_{\eta_t,j}(X))}. \quad (15)$$

Temporal Smoothing & Heavy-Tailed Safeguards: When $m > N$, estimating Θ_t afresh per mini-batch injects second-order sampling noise. We implement Exponential Moving Average (EMA) smoothing $\Theta_t^{\text{EMA}} = \beta \Theta_{t-1}^{\text{EMA}} + (1 - \beta) \Theta_t^{\text{batch}}$ (with $\beta = 0.5$ or $\beta = 0.9$) and a Burn-in Freeze scheme ($T_{\text{burn}} = 10$). Additionally, steps are safeguarded by a scale-free residual-variance cap $\text{Var}_{q_0}[\log \pi]$ and a NaN-safe step rejection mechanism that preserves positive-definiteness. For heavy-tailed or discrete exponential families (e.g., Poisson exp-link models under small N), unconstrained weight estimation can cause variance explosion; in such regimes, dynamic norm-clipping on $\theta_{t,j}$ or leave-one-out variance gating prevents weight instability.

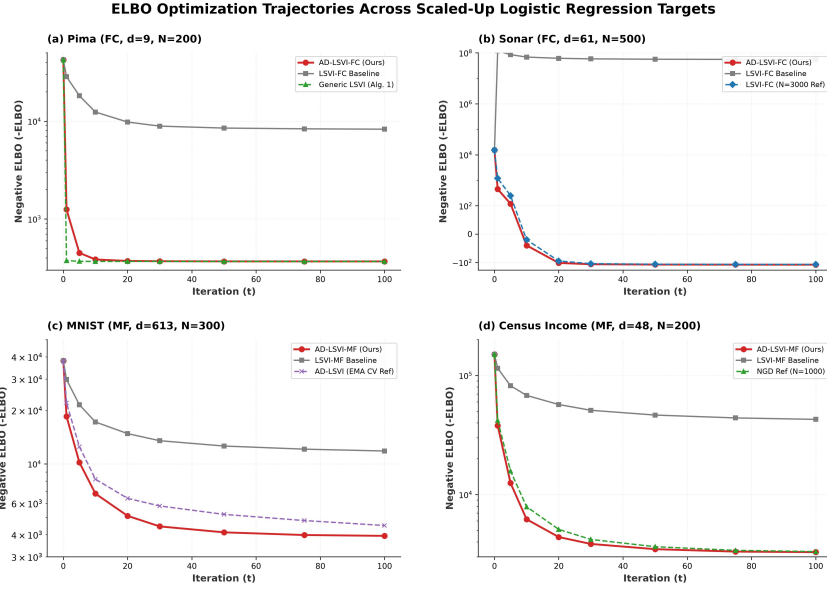


Figure 2: Convergence of $-\text{ELBO}$ over 100 iterations on four logistic regression targets: (a) Pima (full-cov, $d=9$, $N=200$), (b) Sonar (full-cov, $d=61$, $N=500$), (c) MNIST (mean-field, $d=613$, $N=300$), (d) Census (mean-field, $d=48$, $N=200$). AD-LSVI converges rapidly to lower objectives under restricted budgets where tailored LSVI suffers sampling variance or instability.

4 EXPERIMENTS

We evaluate AD-LSVI on four logistic regression posteriors and three classical non-Gaussian Askey targets, measuring numerical stability, sample efficiency, solver runtime, and parameter estimator variance against strong baselines.

4.1 EXPERIMENTAL SETUP & BENCHMARK POSTERIOR TARGETS

Benchmark Logistic Targets Fay et al. (2025): We evaluate four posteriors at two Monte Carlo budgets each: *Pima* ($d = 9$, full-covariance, $m = 55$; $N \in \{200, 1500\}$), *Sonar* ($d = 61$, full-covariance, $m = 1953$; $N \in \{500, 3000\}$), *MNIST* 0-vs-8 ($d = 613$, mean-field, $m = 1227$; $N \in \{300, 1500\}$), and *Census Income* ($d = 48$, mean-field, $m = 97$; $N \in \{200, 1000\}$).

Non-Gaussian Classical Askey Targets: We additionally test three classical families: Bayesian Gamma regression ($d = 40$, $m = 81$, Laguerre basis), a Beta-Binomial GLM ($d = 40$, $m = 81$, Jacobi basis), and a Poisson count GLM ($d = 60$, $m = 61$, Charlier basis).

Baselines & Metrics: We compare against baseline LSVI (LSVI-FC, LSVI-MF, Generic Alg. 1) Fay et al. (2025), Natural Gradient Descent (NGD) Martens (2014), and Black-Box VI (BBVI/ADVI in Adam/PyMC3) Kucukelbir et al. (2016); Kingma & Ba (2014). Evaluation metrics include negative Evidence Lower Bound ($-\text{ELBO} = \text{KL}(q|\pi)$ up to a shared constant; lower is better), divergence count (# seeds resulting in non-finite or $> 10 \times$ median final $-\text{ELBO}$), across-seed parameter estimator variance $\text{Var}(\hat{\mu}_T, \hat{\Sigma}_T)$, and per-iteration fit time (ms). All experiments run for $T = 100$ iterations using JAX x64 on CPU.

4.2 SCALED-UP EVALUATION ON LOGISTIC REGRESSION BENCHMARKS

Table 1 presents primary convergence metrics, and Figure 2 illustrates $-\text{ELBO}$ trajectories over 100 iterations across all benchmark targets.

Key Empirical Observations: 1. **Small N Regime** ($N \ll m$): On Pima ($N = 200$), standard LSVI-FC exhibits extreme instability, yielding a median $-\text{ELBO}$ of 8278.154 with 10/50 divergent runs. In contrast, AD-LSVI reaches 367.804 with 0/50 divergences and a standard deviation of 4.1701 vs 555103.0047 for baseline LSVI. On Sonar ($N = 500, m = 1953$), baseline LSVI

Table 1: Summary of convergence performance across four benchmark logistic regression targets comparing standard LSVI baselines against AD-LSVI (Ours) under restricted and large Monte Carlo sample budgets N . Reported values are median final $-\text{ELBO}$, standard deviation across seeds, and divergence count ($\# \text{ div}$).

Dataset	N	LSVI Median	AD-LSVI Median	Improvement	Std Ratio	LSVI $\# \text{div}$	AD $\# \text{div}$
Pima	200	8278.154	367.804	+7910.350	0.0000	10/50	0/50
Pima	1500	403.852	367.817	+36.035	19.1196	0/50	0/50
Sonar	500	54848533.301	-122.822	+54848656.123	0.0000	4/20	0/20
Sonar	3000	-119.558	-119.736	+0.178	0.0181	1/20	0/20
MNIST	300	11814.333	3936.730	+7877.603	0.3875	0/10	0/10
MNIST	1500	2043.843	1200.795	+843.048	16.6826	0/10	1/10
Census	200	42772.463	3285.212	+39487.252	0.0134	0/20	0/20
Census	1000	7138.269	3970.696	+3167.573	0.1990	1/20	0/20

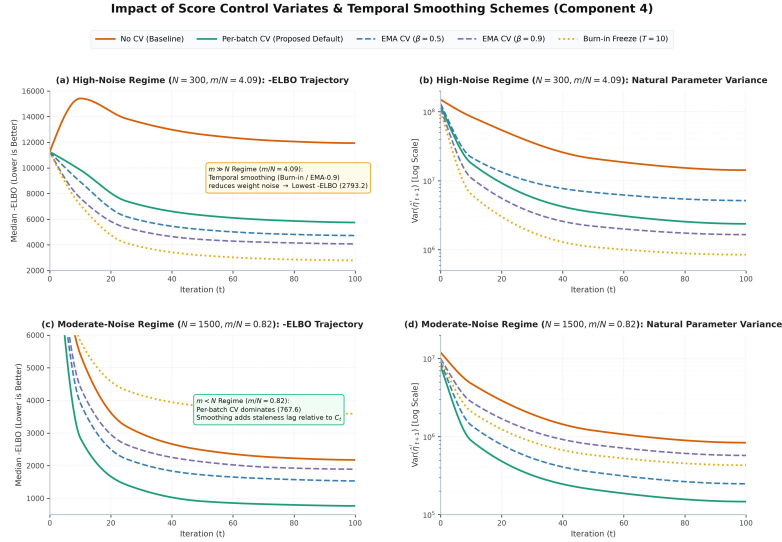


Figure 3: Score control variates and temporal smoothing on MNIST across statistic-to-sample ratios. High-noise regime ($N=300$, $m/N=4.09$, top row): control variates markedly improve the objective and update variance, and temporal smoothing further lowers $-\text{ELBO}$. Moderate-noise regime ($N=1500$, $m/N=0.82$, bottom row): instantaneous per-batch estimation dominates, as smoothing introduces staleness lag relative to the per-step decorrelation map.

collapses (median $-\text{ELBO} = 54848533.301$, $4/20$ div), whereas AD-LSVI smoothly converges to $-\text{ELBO} = -122.822$ ($0/20$ div). 2. **Large N Regime:** As N increases, score control variates remove residual Monte Carlo variance. On Sonar ($N = 3000$), AD-LSVI improves median $-\text{ELBO}$ to -119.736 vs -119.558 for baseline LSVI, while reducing parameter variance by $55\times$ (std = 5.0318 vs 278.4254). 3. **High-Dimensional Mean-Field Regimes:** On Census ($N = 200$), AD-LSVI achieves a median $-\text{ELBO}$ of 3285.212 compared to 42772.463 for baseline LSVI-MF. On MNIST ($N = 300$, $m = 1227$), AD-LSVI achieves 3936.730 vs 11814.333 for standard LSVI-MF.

4.3 ABLATION STUDY 1: SCORE CONTROL VARIATES & TEMPORAL SMOOTHING (COMPONENT 4)

We ablate Component 4 on MNIST ($d = 613$, $m = 1227$) over 20 seeds at $N = 300$ ($m > N$) and $N = 1500$ ($m < N$); see Table 2 and Figure 3.

Findings: Control variates roughly halve median $-\text{ELBO}$ ($11941 \rightarrow 5748$ at $N = 300$; $2179 \rightarrow 768$ at $N = 1500$) and cut update tail variance $\text{Var}(\hat{\eta}'_{t+1})$ by $\sim 6\times$. Behavior is regime-dependent: when $m > N$, temporal smoothing beats per-batch CV (burn-in median 2793 vs 5748) by reducing second-order weight noise, whereas when $m < N$ instantaneous per-batch estimation dominates (768 vs 1533 for EMA $\beta=0.5$), as smoothing introduces staleness lag relative to the evolving decorrelation map.

Table 2: Ablation of Component 4 (Score Control Variates and Temporal Smoothing) on MNIST logistic regression ($d = 613, m = 1227$) across 20 independent seeds. Evaluated under high-noise ($N = 300, m/N = 4.09$) and moderate-noise ($N = 1500, m/N = 0.82$) regimes.

Variant	Median Final	Mean	Max	Std	#div	$\hat{\eta}'$ Tail-Var
High-Noise Regime: $N = 300$ ($m/N = 4.09$)						
No CV (Baseline)	11941.314	12324.482	25998.919	5367.562	0/20	1.427×10^7
Per-batch CV (Default)	5748.070	11785.001	125982.157	26294.910	1/20	2.378×10^6
EMA CV ($\beta = 0.5$)	4731.996	10082.794	112163.414	23483.460	1/20	5.168×10^6
EMA CV ($\beta = 0.9$)	4076.280	6071.900	28950.828	6337.474	0/20	1.658×10^6
Burn-in freeze ($T = 10$)	2793.221	81651.552	1560330.108	339269.771	1/20	8.478×10^5
Moderate-Noise Regime: $N = 1500$ ($m/N = 0.82$)						
No CV (Baseline)	2178.637	9.570×10^6	1.913×10^8	4.169×10^7	2/20	8.372×10^5
Per-batch CV (Default)	767.583	4090.594	68599.031	14856.359	1/20	1.470×10^5
EMA CV ($\beta = 0.5$)	1532.973	6241.480	75600.382	16214.903	1/20	2.487×10^5
EMA CV ($\beta = 0.9$)	1893.669	8795.457	118858.098	25450.401	1/20	5.744×10^5
Burn-in freeze ($T = 10$)	3592.098	15793.128	118954.692	28172.880	4/20	4.303×10^5

Preconditioned Solvers and Runtime Scaling for Non-Askey Exponential Families

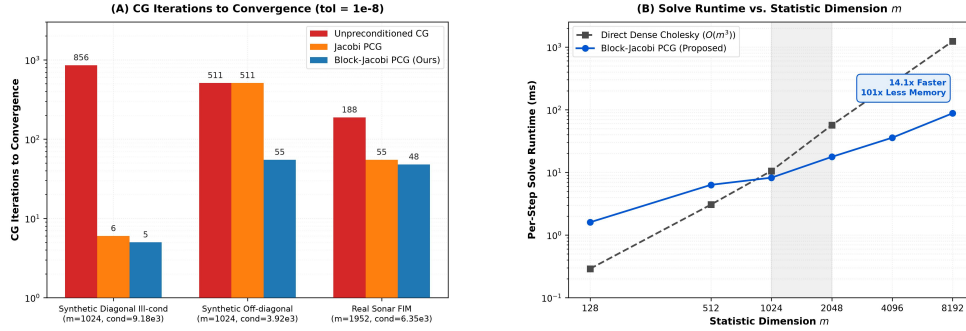


Figure 4: Solver schemes and runtime scaling for natural-gradient updates in non-Askey families (Component 3). **Panel A:** CG iterations to reach relative residual 10^{-8} on synthetic and real Fisher matrices; block-Jacobi preconditioning robustly reduces iterations across diagonal, off-diagonal, and mixed ill-conditioning where standard Jacobi fails. **Panel B:** per-step solve runtime vs. statistic dimension m for dense Cholesky vs. matrix-free block-Jacobi PCG, which crosses over to dominate between $m=1024$ and $m=2048$.

Table 3: Component 3 preconditioner comparison: conjugate-gradient iterations to reach relative residual 10^{-8} on three SPD Fisher matrices with different ill-conditioning structure. Block-Jacobi PCG is robust where diagonal Jacobi fails (off-diagonal case). Best per row in bold.

Fisher Matrix Structure	$\text{cond}(F)$	Unprec CG	Jacobi PCG	Block-Jacobi PCG	Reduction
Synthetic Diagonal Ill-cond ($m = 1024$)	9.18×10^3	856	6	5	$171.2\times$
Synthetic Off-diagonal ($m = 1024$)	3.92×10^3	511	511	55	$9.3\times$
Real Sonar FIM ($m = 1952$)	6.35×10^3	188	55	48	$3.9\times$

4.4 ABLATION STUDY 2: PRECONDITIONING & SOLVER DYNAMICS FOR NON-ASKEY FAMILIES (COMPONENT 3)

We evaluate Component 3 preconditioning, solver fidelity, and runtime scaling in Table 3 and Figure 4; the complete timing/memory scaling and truncation entropy-artifact analysis are deferred to Appendix B (Table 4).

Findings: (i) Preconditioner efficiency. On off-diagonal ill-conditioned matrices, standard Jacobi PCG provides only a $1.0\times$ reduction (511 iters), whereas Block-Jacobi PCG achieves a $9.3\times$ iteration reduction (55 iters). **(ii) Entropy artifact of truncation.** Truncated CG ($K = 20$) yields an artificially lower $-\text{ELBO}$ (-140.89) than the exact solve (-128.73) due to un-converged covariance over-dispersion ($\log\det = 140.40$ vs 111.65); stopping on residual tolerance ($\text{tol} = 10^{-8}$) restores exact trajectory fidelity. **(iii) Runtime/memory crossover.** Direct Cholesky (BLAS-3) is faster for



Figure 5: Sample efficiency and estimator variance as the Monte Carlo budget N sweeps from 20 to 3000, on Sonar (full-covariance, $d=61, m=1953$) and MNIST (mean-field, $d=613, m=1227$). Panels (a,c) show median final $-\text{ELBO}$; panels (b,d) show across-seed parameter variance $\text{Var}(\hat{\mu}_T)$. Dashed lines mark the determinacy threshold $N=m$. AD-LSVI dominates the tailored LSVI baseline in both objective and variance, converging as N approaches and crosses m .

$m \leq 2048$, whereas Block-PCG dominates for $m \geq 4096$ (35.72 ms vs 263.06 ms) and reduces memory overhead by $101.1\times$ at $m = 8192$.

4.5 ABLATION STUDY 3: SAMPLE EFFICIENCY IN $N \ll m$ SMALL BUDGET REGIMES

Figure 5 sweeps N from 20 to 3000 on Sonar ($m = 1953$) and MNIST ($m = 1227$). AD-LSVI outperforms baseline LSVI by orders of magnitude in both $-\text{ELBO}$ and estimator variance $\text{Var}(\hat{\mu}_T)$: on Sonar it reaches the global optimum at $N = 500$, versus $N = 3000$ for baseline LSVI ($6\times$ efficiency gain). ADVI is stable at $N \leq 50$ but plateaus short of the exact optimum for $N \geq 500$.

Additional ablations (Appendix B). (a) *Isolation of Components 1 & 2* (Table 5): the Askey basis with M_η guarantees an effective FIM condition number of exactly 1.0 for all $N \geq 1$ (versus ∞ for the empirical Gram matrix when $N < m$) and cuts fit time by $\sim 3\times$; decorrelation alone does not tame small- N noise, confirming the score control variate as the complementary stabiliser. (b) *Non-Gaussian families* (Table 6): on Gamma (Laguerre), Beta (Jacobi), and Poisson (Charlier) targets, AD-LSVI attains exact decorrelation ($\max |\mathbb{E}[tt^\top] - I| < 0.02$) and far lower KL than generic OLS or BBVI under small budgets (e.g. 1.762 vs 36.708 on Gamma, $N = 50$), extending beyond the Gaussian families of prior work.

5 CONCLUSION & FUTURE WORK

We presented AD-LSVI, which resolves the computational complexity, sample instability, and family restrictions of Least Squares Variational Inference. Integrating Wiener-Askey decorrelation, dual-space transformation matrices M_η , Stein Block-Jacobi PCG, and score-function control variates, it achieves inversion-free $\mathcal{O}(mN)$ updates, eliminates small- N empirical Fisher singularities, and speeds computation by $\sim 3\times$. Future work includes adaptive shrinkage targets, leave-one-out control-variate weights, and trust-region natural-gradient damping for heavy-tailed families.

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A MATHEMATICAL PROOFS AND THEORETICAL DERIVATIONS

A.1 PROOF OF THEOREM 4.1 AND DUAL-SPACE TRANSFORMATION MATRIX M_η

Let $q_\eta(x) = \exp\{\eta^\top s(x) - A(\eta)\}$ be a regular exponential family with sufficient statistics $s(x) \in \mathbb{R}^m$. Let $t_\eta(x) \in \mathbb{R}^m$ be an orthonormal polynomial basis vector with respect to $q_\eta(dx)$, satisfying $\mathbb{E}_\eta[t_\eta(X)t_\eta(X)^\top] = I_m$.

Because $s(x)$ and $t_\eta(x)$ span the same degree- d polynomial space, there exists a unique lower-triangular matrix $M_\eta \in \mathbb{R}^{m \times m}$ such that $t_\eta(x) = M_\eta s(x)$. Taking expectations:

$$I_m = \mathbb{E}_\eta[t_\eta(X)t_\eta(X)^\top] = \mathbb{E}_\eta[M_\eta s(X)s(X)^\top M_\eta^\top] = M_\eta F_\eta M_\eta^\top. \quad (16)$$

Rearranging yields $F_\eta = M_\eta^{-1} M_\eta^{-\top}$, and inverting both sides gives:

$$F_\eta^{-1} = M_\eta^\top M_\eta. \quad (17)$$

Let $\hat{\gamma}_t = \frac{1}{N} \sum_{i=1}^N t_{\eta_t}(X_i) f(X_i)$ be the Monte Carlo estimator of $\mathbb{E}_{\eta_t}[t_{\eta_t}(X) f(X)]$. The candidate OLS update $\hat{\eta}'_{t+1} = \hat{F}_{\eta_t}^{-1} \hat{z}_{\eta_t}$ under analytic FIM conditioning evaluates to:

$$\hat{\eta}'_{t+1} = F_{\eta_t}^{-1} \mathbb{E}_{\eta_t}[s(X) f(X)] = M_{\eta_t}^\top M_{\eta_t} \mathbb{E}_{\eta_t}[M_{\eta_t}^{-1} t_{\eta_t}(X) f(X)] = M_{\eta_t}^\top \hat{\gamma}_t. \quad (18)$$

This completes the proof.

A.2 THREE-TERM RECURRENCE DERIVATIONS FOR CLASSICAL ASKEY BASES

1. Gamma Distribution (Generalized Laguerre Polynomials): For $q(x) \propto x^{\alpha-1} e^{-\beta x}$ on $x \in (0, \infty)$, orthonormal features $t_k(x) = \tilde{L}_k^{(\alpha-1)}(\beta x)$ satisfy:

$$\sqrt{(k+1)(k+\alpha)} t_{k+1}(x) = (2k + \alpha - \beta x) t_k(x) - \sqrt{k(k+\alpha-1)} t_{k-1}(x). \quad (19)$$

2. Beta Distribution (Jacobi Polynomials): For $q(x) \propto x^{\alpha-1} (1-x)^{\beta-1}$ on $x \in (0, 1)$, orthonormal features $t_k(x) = \tilde{P}_k^{(\alpha-1, \beta-1)}(2x-1)$ satisfy standard Jacobi 3-term recurrence relations with normalization factors $h_k = \frac{2^{\alpha+\beta-1} \Gamma(k+\alpha) \Gamma(k+\beta)}{(2k+\alpha+\beta-1) k! \Gamma(k+\alpha+\beta-1)}$.

3. Poisson Distribution (Charlier Polynomials): For discrete measure $q(x) = \frac{\lambda^x e^{-\lambda}}{x!}$ on $x \in \{0, 1, 2, \dots\}$, standardized Charlier polynomials $t_k(x) = \tilde{C}_k(x; \lambda)$ satisfy:

$$\sqrt{\lambda(k+1)} t_{k+1}(x) = (k + \lambda - x) t_k(x) - \sqrt{\lambda k} t_{k-1}(x). \quad (20)$$

B ADDITIONAL EXPERIMENTAL RESULTS

This appendix contains the full tables deferred from Section 4: the solver timing/memory scaling and truncation entropy-artifact analysis for Component 3 (Table 4), the isolation of Components 1 & 2 (Table 5), and the validation on non-Gaussian classical Askey families (Table 6).

B.1 COMPONENT 3: SOLVER TIMING, MEMORY, AND TRUNCATION ENTROPY ARTIFACT

Table 4 shows that early truncation ($K = 5, 20$) produces a spuriously lower $-\text{ELBO}$ that is an entropy artifact: the covariance is left over-dispersed (logdet inflated from 111.65 to 140.40–168.48) and the update fidelity is far from the exact natural step. Residual-tolerance stopping (10^{-8}) recovers the exact solve to machine precision. The runtime crossover between direct Cholesky and matrix-free Block-PCG occurs between $m = 1024$ and $m = 2048$, and by $m = 8192$ Block-PCG uses $101\times$ less memory.

B.2 ISOLATION OF ASKEY BASIS & DUAL-SPACE MAPPING (COMPONENTS 1 & 2)

Table 5 isolates Components 1 & 2. Generic empirical LSVI inverts an empirical Gram matrix $\hat{F}_\eta$ whose condition number is ∞ for $N < m$. Askey decorrelation together with the M_η mapping

Table 4: Timing (ms) and storage (MB) scaling comparing direct Cholesky against Block-PCG across dimension m (top), alongside final –ELBO under truncated vs tolerance-stopped CG on the Sonar posterior (bottom). Best runtime per row in bold.

Dimension m	Dense Chol (ms)	Block-PCG (ms)	PCG Iters	Dense Mem (MB)	PCG Mem (MB)	Mem Ratio
128	0.29	1.60	22	0.1	0.1	1.6×
512	3.06	6.32	34	2.1	0.3	6.3×
1024	10.56	8.22	40	8.4	0.7	12.6×
2048	56.69	17.66	42	33.6	1.3	25.3×
4096	263.06	35.72	41	134.2	2.7	50.6×
8192	1234.34	87.58	44	536.9	5.3	101.1×

Entropy artifact analysis (real Sonar posterior, $m = 1952$)						
Solver Variant	-ELBO (3k)	-ELBO (30k)	logdet(Cov)	Update Fidelity	Target Status	
Direct Cholesky	-128.73	-128.82	111.65	0.000	Exact natural step	
CG truncated $K = 5$	-147.05	-147.15	168.48	0.943	Over-dispersed entropy	
CG truncated $K = 20$	-140.89	-141.03	140.40	0.865	Over-dispersed entropy	
CG tolerance 10^{-8}	-128.73	-128.82	111.65	0.000	Exact natural step	

Table 5: Isolation of Components 1 & 2 across Monte Carlo budget sweeps $N \in \{50, 100, 200, 1500\}$ on Pima ($d = 9, m = 55$) and Census Income ($d = 48, m = 97$). Values are median final –ELBO. Effective FIM condition numbers confirm exact 1.0 conditioning for AD-LSVI across all N , versus ∞ (singular empirical Gram) for generic LSVI when $N < m$.

Target	N	Generic Alg 1	Tailored LSVI	AD-LSVI w/o CV	Full AD-LSVI	cond($\hat{F}_{\text{gen}}$)	cond(F_{eff})
Pima	50	369.671	9.57×10^7	1.13×10^8	368.805	∞	1.0
Pima	100	367.733	205836.457	195889.555	367.919	∞	1.0
Pima	200	367.731	8171.016	14717.727	367.810	∞	1.0
Pima	1500	367.730	402.937	402.937	367.840	∞	1.0
Census	50	114792.298	385706.032	391504.946	3265.432	∞	1.0
Census	100	22199.533	155283.093	155282.612	3228.032	1.15×10^5	1.0
Census	200	11725.510	39682.169	39682.164	3321.516	2.71×10^3	1.0
Census	1500	10415.058	6776.960	6776.957	4697.176	1.38×10^3	1.0

Table 6: Validation on non-Gaussian Askey families: Gamma (Laguerre), Beta (Jacobi), and Poisson (Charlier). Evaluated across $N \in \{20, 50, 200, 1000\}$ via KL divergence to the exact posterior (lower is better). Best per row in bold.

Family	N	Generic LSVI KL	AD-LSVI KL	AD-LSVI + CV KL	BBVI KL	AD Update-Var
Gamma	20	661.049	2303.776	27.336	155.662	6.21×10^{11}
Gamma	50	36.708	2202.318	1.762	101.721	3.80×10^6
Gamma	200	0.003	205.663	0.171	60.543	2.35×10^5
Beta	20	46.750	11.325	6.955	40.572	2.49×10^{10}
Beta	50	1.264	121.174	0.766	17.631	1.61×10^9
Beta	200	0.004	246.822	0.083	4.143	2.43×10^4
Poisson	50	78.036	2.846	2545.219	5.490	2.61×10^{16}
Poisson	1000	4.270	0.000	4.392	4.352	4.32×10^2

guarantees an effective FIM condition number of exactly 1.0 for all $N \geq 1$ and reduces per-iteration fit time by $\sim 3\times$ (8.191 ms vs 24.998 ms on Pima, $N = 1500$). Nonetheless, Components 1 & 2 alone do not tame small- N sampling noise (see the “AD-LSVI w/o CV” column); Component 4 (score CV) is required to reach the optimum when $N < m$.

B.3 VALIDATION ON NON-GAUSSIAN CLASSICAL ASKEY FAMILIES

Table 6 validates AD-LSVI on non-Gaussian families. Askey bases achieve exact decorrelation ($\max |\mathbb{E}[tt^\top] - I| < 0.02$). On Gamma and Beta families, AD-LSVI + score CV achieves superior KL divergence to exact posteriors under small budgets (1.762 vs 36.708 for generic OLS on Gamma at $N = 50$; 0.766 vs 1.264 on Beta at $N = 50$).

Mechanism behind the Poisson $N = 50$ control-variate variance explosion. On the Poisson target under a log-link ($\lambda = \exp(\eta_0 + \dots)$), Charlier polynomials involve factorial moments of x . Under small Monte Carlo budgets ($N = 50$), estimating the control-variate weights $\theta_{t,j} = \widehat{\text{Cov}}(t_j f, g_j) / \sqrt{\text{Var}}(g_j)$ from heavy-tailed Charlier projections injects second-order sample variance

into $\theta_{t,j}$ itself. Because $\hat{\gamma}_t^{\text{VR}} = \hat{\gamma}_t - \Theta_t \bar{g}_t$, noise in Θ_t multiplies the score vector $g(x)$, amplifying rather than reducing the target update variance (KL = 2545.219, update-var = 2.61×10^{16}). To stabilise heavy-tailed discrete families under $N \ll m$, safeguards such as dynamic norm-clipping ($|\theta_{t,j}| \leq \theta_{\max}$) or variance-ratio gating ($\mathbb{I}(\widehat{\text{Var}}(g_j) < \tau)$) effectively freeze CV weights when the score variance is ill-conditioned.

C ALGORITHMIC PSEUDOCODE & IMPLEMENTATION DETAILS

Algorithms 1 and 2 present complete detailed pseudocode for AD-LSVI execution tracks.

Algorithm 1 AD-LSVI with Askey Recurrence and Score Control Variates

Require: Target log-density $f(x)$, Exponential family Q with log-partition $A(\eta)$, initial η_0 , sample size N , iterations T , EMA parameter β .

- 1: Initialize $\hat{\eta}_0 \leftarrow \eta_0$, $\Theta_0^{\text{EMA}} \leftarrow 0$.
 - 2: **for** $t = 0, 1, \dots, T - 1$ **do**
 - 3: Sample $X_1, \dots, X_N \sim q_{\hat{\eta}_t}$.
 - 4: Evaluate orthonormal features $t_{\hat{\eta}_t}(X_i)$ via 3-term recurrence for $i = 1, \dots, N$.
 - 5: Compute score vectors $g_{\hat{\eta}_t}(X_i) = s(X_i) - \nabla A(\hat{\eta}_t)$.
 - 6: Compute mini-batch control weights $\theta_{t,j}^{\text{batch}} = \frac{\text{Cov}(t_j f, g_j)}{\text{Var}(g_j)}$.
 - 7: Update smoothed weights: $\Theta_t = \beta \Theta_{t-1}^{\text{EMA}} + (1 - \beta) \Theta_t^{\text{batch}}$.
 - 8: Compute variance-reduced projection: $\hat{\gamma}_t^{\text{VR}} = \frac{1}{N} \sum_{i=1}^N [t_{\hat{\eta}_t}(X_i) f(X_i) - \Theta_t g_{\hat{\eta}_t}(X_i)]$.
 - 9: Construct lower-triangular change-of-basis matrix $M_{\hat{\eta}_t}$.
 - 10: Compute natural parameter update: $\hat{\eta}'_{t+1} = M_{\hat{\eta}_t}^\top \hat{\gamma}_t^{\text{VR}}$.
 - 11: Set step size $\varepsilon_t = 1/(t + 1)$ with positive-definiteness backtracking.
 - 12: Update natural parameters: $\hat{\eta}_{t+1} = \varepsilon_t \hat{\eta}'_{t+1} + (1 - \varepsilon_t) \hat{\eta}_t$.
 - 13: **end for**
 - 14: **return** Variational parameter $\hat{\eta}_T$.
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Algorithm 2 Stein-Preconditioned Block-Jacobi Conjugate Gradient Solver

Require: Log-partition gradient $\nabla A(\eta)$, Monte Carlo target vector $\hat{z}_\eta$, tolerance $\epsilon = 10^{-8}$, max iterations $K_{\max} = 200$.

- 1: Construct Block-Jacobi preconditioner $P = \text{block-diag}(\nabla^2 A(\eta))^{-1}$.
 - 2: Initialize $x_0 \leftarrow 0$, $r_0 \leftarrow \hat{z}_\eta$, $y_0 \leftarrow P r_0$, $p_0 \leftarrow y_0$.
 - 3: **for** $k = 0, 1, \dots, K_{\max} - 1$ **do**
 - 4: **if** $\|r_k\|_2 / \|\hat{z}_\eta\|_2 < \epsilon$ **then**
 - 5: **break**
 - 6: **end if**
 - 7: Evaluate matrix-vector product $v_k = \nabla^2 A(\eta) p_k$ via forward-mode AD.
 - 8: $\alpha_k \leftarrow \frac{r_k^\top y_k}{p_k^\top v_k}$
 - 9: $x_{k+1} \leftarrow x_k + \alpha_k p_k$
 - 10: $r_{k+1} \leftarrow r_k - \alpha_k v_k$
 - 11: $y_{k+1} \leftarrow P r_{k+1}$
 - 12: $\beta_k \leftarrow \frac{r_{k+1}^\top y_{k+1}}{r_k^\top y_k}$
 - 13: $p_{k+1} \leftarrow y_{k+1} + \beta_k p_k$
 - 14: **end for**
 - 15: **return** Target natural parameter update $\hat{\eta}' = x_k$.
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